

Android TV Remote

Features and Development Hurdles

Project: AndroidTVRemote | Date: June 30, 2026

1. App Overview

WiFi smart remote and casting app for Android TV / Google TV using protocol v2 (TLS ports 6466/6467) plus Google Cast. Built in Kotlin with MVVM: remotecontrol library module + app UI module.

2. Features

2.1 Settings - TV Connection

- mDNS scan for Android TVs on WiFi
- Tap TV to pair; 6-char code entry; Complete Pairing button
- Manual IP fallback; Get new code; Reconnect without re-pairing
- Saved pairing state; phone IP display; live status labels

2.2 Remote Tab

- D-pad, Home, Back, Power, Apps, TV Input, Voice Search
- Volume, Channel, Mute, Play/Pause, Rewind, Forward
- Shortcuts: Netflix, YouTube, Prime Video, Hotstar
- Connection status; jump to Cast tab

2.3 Cast Tab

- Google Cast device picker (MediaRouter)
- Cast photos (Downloads/Gallery), videos, audio
- Screen mirror via MediaProjection + MJPEG HTTP stream
- Foreground service + notification while mirroring

2.4 Session & Architecture

- Remote pauses during cast/mirror; auto-resumes after
- MVVM: ViewModels, Repositories, AppContainer DI
- StateFlow UI; ViewBinding; glass-style bottom nav UI

2.5 Reliability

- SafeRun error wrapping; OperationResult for cast
- CrashGuard logging; safe fragments and NSD/Cast handling

3. Modules

- app: ui/, data/, di/, util/
- remotecontrol: PairingManager, RemoteManager, TLS, certs

4. Development Hurdles

Pairing & UX

Submit button / pair loop

Problem: Code on TV but no submit; Pair again re-sent code.

Solution: Complete Pairing + waitingForCode guard.

Manual IP first

Problem: Users wanted tap-to-pair from scan list.

Solution: TV cards primary; manual IP optional.

Re-pair after cast

Problem: Cast broke remote; endless new pairing codes.

Solution: PairingStore, connect vs pair, ConnectionCoordinator pause/resume.

Threading & Lifecycle

Wrong thread UI

Problem: Callbacks updated UI off main thread.

Solution: mainHandler post on all manager callbacks.

Binding NPE

Problem: Tab switch crashed on null binding.

Solution: `_binding` guards; `repeatOnLifecycle`.

Fragment transaction crash

Problem: commit during state save.

Solution: `isStateSaved` / `isDestroyed` checks.

Build & Code

Windows path too long

Problem: Gradle transform `MAX_PATH` error.

Solution: Short build dir in `build.gradle.kts`.

combine() 5-flow limit

Problem: `SettingsViewModel` compile error.

Solution: Nested combine for 6+ flows.

MVVM refactor duplicates

Problem: Old root files caused duplicate class errors.

Solution: Deleted legacy files; manifest path updates.

Cast vs Remote

Protocols conflict

Problem: Cast dropped remote socket.

Solution: Separate session modes; pause remote on cast.

Auto-pair fallback

Problem: 8s timeout re-paired already-paired TVs.

Solution: Fallback only on explicit pair; reconnect backoff.

Stability

Unhandled exceptions

Problem: NSD, Cast, projection could crash.

Solution: `SafeRun`, `OperationResult`, protocol try/catch.

5. Test Checklist

- Pair, remote keys, tab switch during scan
- Cast photo/video; mirror start/stop; remote pause/resume
- Reconnect paired TV without new code
- Deny permissions â no crash

6. Limitations

- Same WiFi required; emulator cannot reach real TV
- Mirror quality depends on network (MJPEG over HTTP)
- Cast and Remote are separate protocols, coordinated in app